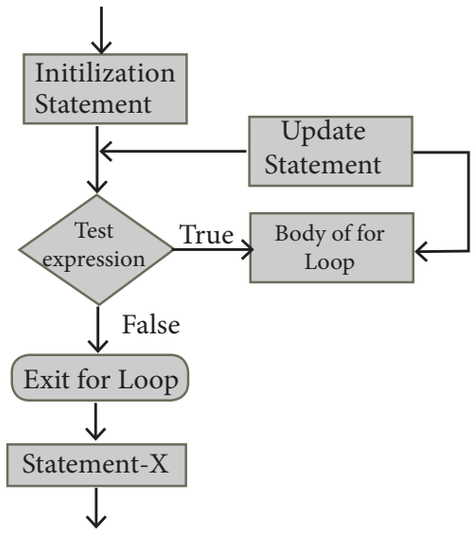
The initialization part is used to initialize variables or declare variable which are executed only once, then the control passes to test-expression. After evaluation of test-expression, if the result is false, the control transferred to statement-x. If the result is true, the body of the for loop is executed, next the control is transferred to update expression. After evaluation of update expression part, the control is transferred to the test-expression part. Next the steps 3 to 5 is repeated. The workflow of for loop and flow chart are shown below.



Flowchart 10.6: Workflow of for loop and flow chart

Illustration 10.7 C++ program to display numbers from 0 to 9 using for loop

```
#include <iostream>
using namespace std;
int main ()
{
int i;
for(i = 0; i< 10; i ++ )
        cout<< "value of i : " <<i<<endl;
return 0;
}
```

Output

```
value of i : 0
value of i : 1
value of i : 2
value of i : 3
value of i : 4
value of i : 5
value of i : 6
value of i : 7
value of i : 8
value of i : 9
```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